

Punna Bhargavi

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of project experience in Generative AI, LLM applications, Agentic AI workflows, NLP pipelines, and REST APIs. Proficient in Python, FastAPI, PostgreSQL, and AWS; shipped **5+** ML solutions across multimodal AI, RAG systems, and automated reporting with documented, quantified outcomes.

EDUCATION

Vardhaman College of Engineering

Shamshabad, TS

B.Tech – Computer Science & Engineering (AI & ML) | **9.61 CGPA**

Aug 2023 – May 2027

- **Coursework:** DSA, ML, DBMS, OS, Computer Networks, OOP, DL, Data Science, Data Visualisation.

EXPERIENCE

Data Analyst & AI Engineer

Jan 2026 – Mar 2026

Vistara Vidyaanidhi Educational Trust

Remote

- Processed **10K+** donation and beneficiary records using Python and PostgreSQL; introduced targeted SQL views and anomaly-detection queries, cutting manual analysis time by **60%** across 3 monthly reporting cycles.
- Spearheaded an Agentic AI reporting system using LLM-based summarization, eliminating **5+ hours** of weekly manual effort for a 4-member admin team; constructed automated data validation checks improving report accuracy by **25%**.

MassMutual India – Full Stack Case Study (AIDEA Program)

Nov 2025

AI/ML Developer | Virtual Internship

Remote

- Engineered an AI-powered recruitment platform with NLP resume parsing and candidate profiling; raised screening throughput by **40%** across **500+** applicant profiles.
- Formulated RESTful APIs and PostgreSQL backend modules; instituted real-time analytics dashboards that cut reporting turnaround by **35%**; refined recruiter workflows reducing shortlisting effort by **30%**.

PROJECTS

Android Malware Detection with Explainable AI | Python, XGBoost, PCA, WOA, SHAP, Plotly

Feb 2026

- Pioneered a malware detection pipeline across **6+ ML algorithms**; applied PCA for **40% dimensionality reduction** and WOA-tuned XGBoost, boosting detection by **15%** and reaching **97% accuracy**.
- Deployed SHAP-based interpretability across the full pipeline, clarifying feature impact for model decisions and improving explainability coverage across all evaluated classifiers.

FloatChat: AI-Driven ARGO Ocean Data Interface | Python, PostGIS, LangChain, RAG, ChromaDB, Plotly

Sep 2025

- Engineered a RAG-backed conversational interface for NetCDF oceanographic data; architected an ingestion pipeline processing **15M+ data points** from **100+ ARGO files** with PostGIS spatial integration.
- Indexed **500K+ metadata records** into ChromaDB for semantic search, attaining **85% LLM query accuracy** with interactive Plotly visualizations – **Smart India Hackathon 2025** submission.

AI Content Suite: Chatbot & PowerPoint Generator | Python, OpenAI API, NLP, FastAPI, Automation

Jul 2025

- Programmed a multimodal LLM chatbot with speech-to-text, text-to-speech, PDF parsing, and image queries via OpenAI API; raised response accuracy by **30%** across **1,000+** test queries on a FastAPI backend.
- Directed a Generative AI pipeline for structured slide generation from natural language prompts, producing **10–15 slide decks** and cutting authoring time by **70%** across **50+** outputs.

Heart Disease Prediction System | Python, scikit-learn, Plotly, Pandas, NumPy

Oct 2025

- Scrutinized **5+** classifiers via feature engineering and cross-validation, achieving **75–90% accuracy**; constructed Plotly dashboards to visualize patient distributions and benchmark model performance.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

Databases & Cloud: PostgreSQL, PostGIS, MySQL, AWS (EC2, S3), Data Pipelines

AI / ML: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Agentic AI, RAG, Prompt Engineering, Explainable AI, Fine-Tuning, Model Evaluation

Frameworks & Libraries: FastAPI, Flask, TensorFlow, scikit-learn, Hugging Face, LangChain, ChromaDB, OpenAI API, OpenRouter API, Pandas, NumPy, Plotly, Streamlit

Tools & Practices: Git, GitHub, VS Code, Cursor, Jupyter Notebook, REST APIs, OOP, SDLC, DSA, CI/CD, Agile

CERTIFICATIONS & ACHIEVEMENTS

Industry & AI: Salesforce Certified AgentForce Specialist | Machine Learning – Udemy | Python – HackerRank | Java – HackerRank

Cloud & Data: AWS Generative AI Foundations | AWS Cloud Foundations | AWS Data Engineering | CCNA: Introduction to Networks

Achievements: **Winner** – Innovators Conclave 2025, BITS Pilani Hyderabad | **Runner-up** – Salesforce AgentForce Hackathon 2026 (24 hrs) | Multiple **1st Place** wins across AI/ML and Full-Stack intra-college hackathons